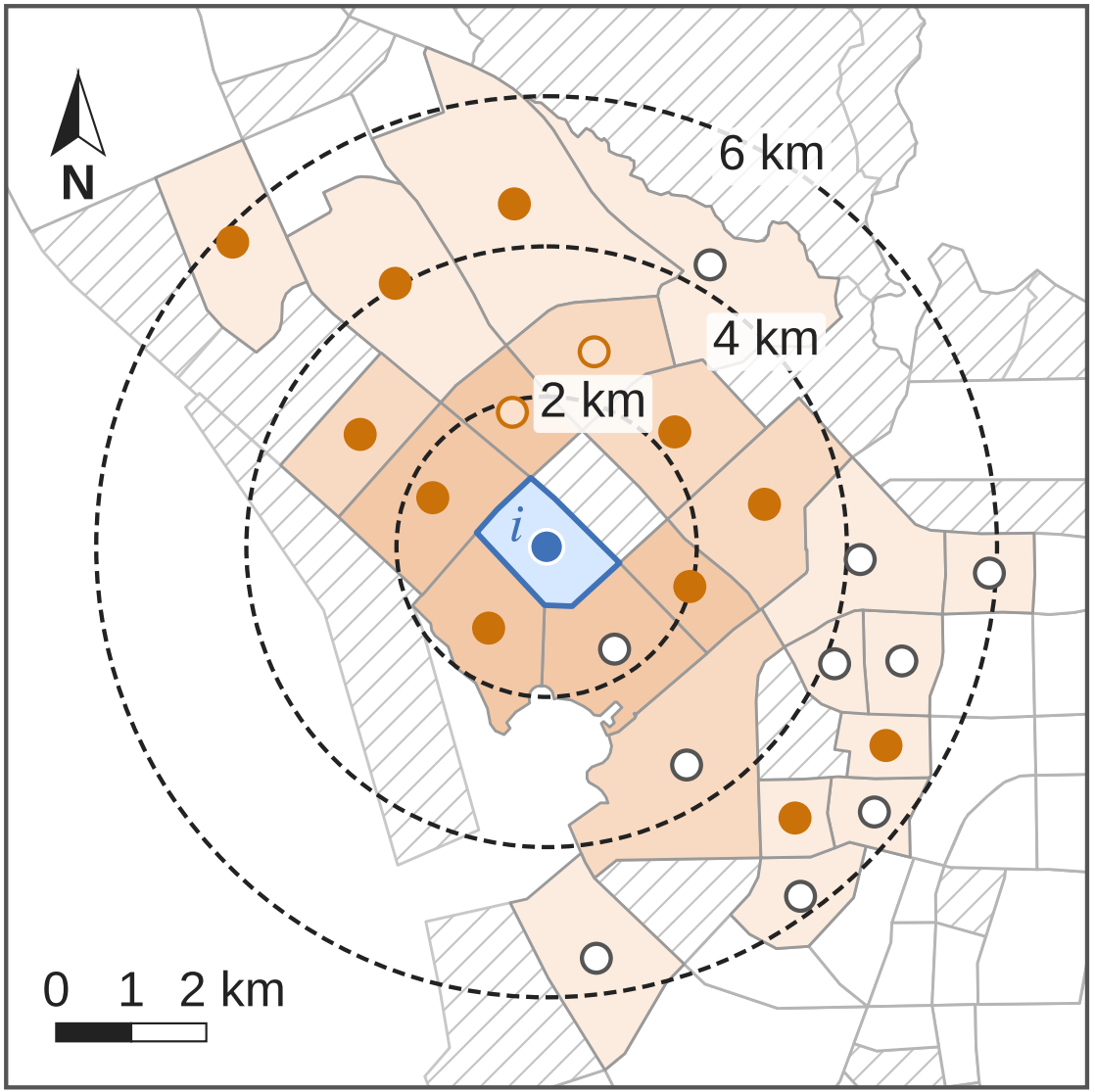
Distance rings around a zone



zone i

ring 1: [0, 2) km, 5 zones

ring 2: [2, 4) km, 5 zones

ring 3: [4, 6) km, 13 zones

zones beyond 6 km

no charging data

time-of-use

weakly varying

fixed price ($\Delta \ell_j \equiv 0$)

Exposure of zone i to ring k:

$$S_{i,t}^{(k)} = \text{mean}\{\Delta \ell_{j,t} : j \in \mathcal{R}_k(i)\}$$

$$\mathcal{R}_k(i) = \{j \neq i : r_{k-1} \leq d_{ij} < r_k\}$$

d_{ij}: distance between centroids